

Clustering and Differential Analysis

Yongjin Park

05 March 2026

Roadmap: From biology to methodology

14: Biological Networks

- Protein-protein, genetic interactions
- Cell-cell interaction networks
- Network as a tool: propagation

15: Clustering

- Graph-based clustering
- Single-cell example
- When clustering goes wrong

Today's lecture

- 1 Graph-based clustering
- 2 Single-cell clustering in practice
- 3 When clustering goes wrong
- 4 Discussion: How should we test cell-cell communication?
- 5 Appendix

Last time: networks encode relationships

- We learned how to build and visualize networks
- Protein-protein interactions, genetic interactions, cell-cell communication
- Today: how do we find **groups** within a network?

Recap: igraph in R

```
library(igraph)
```

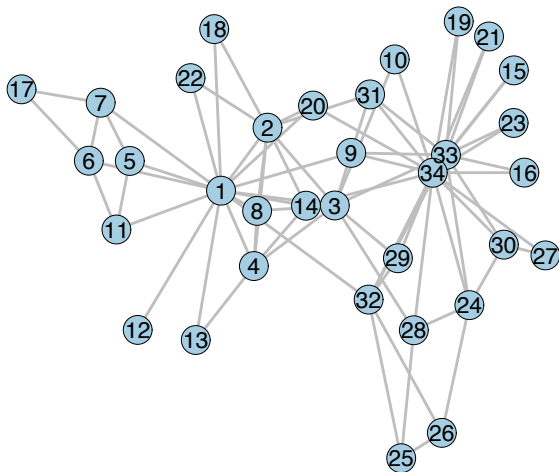
```
karate <- .read.online("http://www-personal.umich.edu/~mejn/netdata/karate.zip")
karate
```

```
## IGRAPH cdaef9d U--- 34 78 --
## + attr: id (v/n)
## + edges from cdaef9d:
## [1] 1-- 2 1-- 3 2-- 3 1-- 4 2-- 4 3-- 4 1-- 5 1-- 6 1-- 7 5-- 7
## [11] 6-- 7 1-- 8 2-- 8 3-- 8 4-- 8 1-- 9 3-- 9 3--10 1--11 5--11
## [21] 6--11 1--12 1--13 4--13 1--14 2--14 3--14 4--14 6--17 7--17
## [31] 1--18 2--18 1--20 2--20 1--22 2--22 24--26 25--26 3--28 24--28
## [41] 25--28 3--29 24--30 27--30 2--31 9--31 1--32 25--32 26--32 29--32
## [51] 3--33 9--33 15--33 16--33 19--33 21--33 23--33 24--33 30--33 31--33
## [61] 32--33 9--34 10--34 14--34 15--34 16--34 19--34 20--34 21--34 23--34
## [71] 24--34 27--34 28--34 29--34 30--34 31--34 32--34 33--34
```

- $n = |V|$: number of vertices, $m = |E|$: number of edges
- d_v : degree of vertex v (how many neighbours?)

A classic example: Karate club (Zachary 1977)

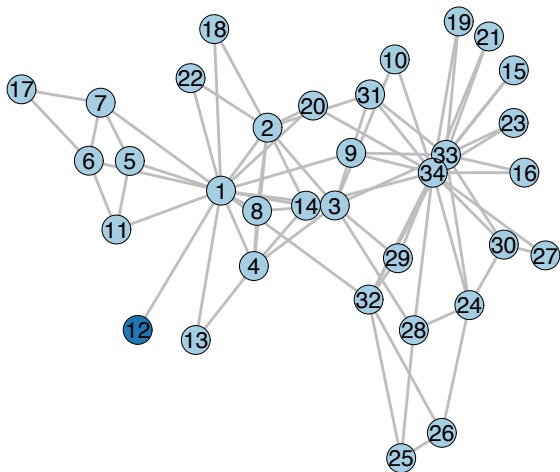
- 34 members, edges = friendships
- The club split into two factions after a dispute
- Can we recover this split from the network alone?
- Who are the “influencers”?
 - **1**: the instructor (“Mr. Hi”)
 - **34**: the president (John A.)



Idea: split the network by cutting edges

- Consider two groups — find a set of edges to cut so the graph splits into two disconnected components
- Min-cut:** the fewest edges to remove

A trivial solution?



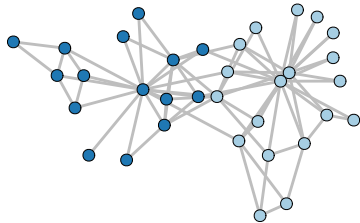
Split G by the fewest edges between two parties

```
.cuts <-  
  as.directed(karate) %>%  
  st_min_cuts(source = "1", target = c("33", "34"))  
.cuts$value / 2  # number of edges to cut
```

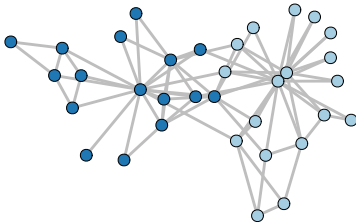
```
## [1] 5
```

Min-cut partitions from source to target

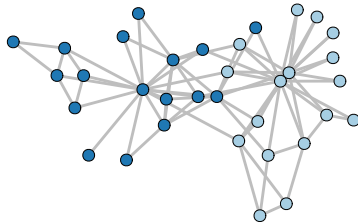
Partition 1 (15 members)



Partition 2 (16 members)



Partition 3 (17 members)



- Multiple solutions with the same min-cut = 5 edges
- But min-cut requires knowing **source and target** — can we find groups without labels?

The network as an adjacency matrix A



- Row/column: a member
- Non-zero element: friendship
- Can you see group structures?

$$A_{ij} = \begin{cases} 1 & i, j \text{ connected} \\ 0 & \text{otherwise} \end{cases}$$

Stochastic block model: a generative model of A ?

- An adjacency matrix A can be **data**

$$A_{ij} \sim \text{Bernoulli} \left(\sum_{k,k'} \theta_{kk'} Z_{ik} Z_{jk} \right)$$

where

$$Z_{ik} = \begin{cases} 1 & i \in \text{cluster/block/membership } k \\ 0 & \text{otherwise} \end{cases}$$

and

$$\theta_{kk'} = P(A = 1 | \text{between communities } k \text{ and } k')$$

- **Goal:** factorize $A \approx Z^T \Theta Z$

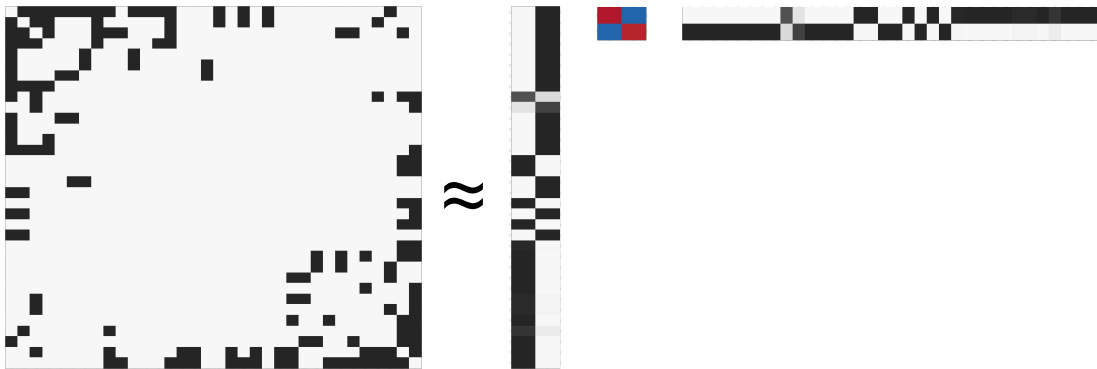
Fitting the stochastic block model

```
library(hsblock)
.out <- fit.hsblock(A, "bernoulli", vbiter=100,
                    inner.iter=10, verbose=FALSE)
Z <- as.matrix(.out$Z)
```

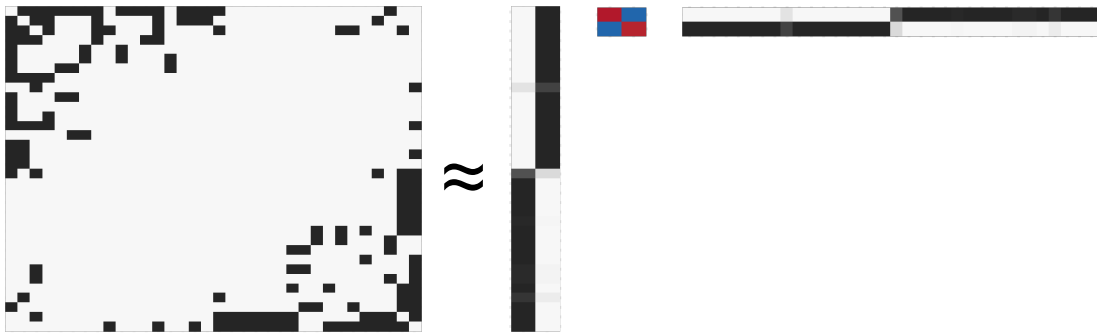
```
n <- ncol(Z)
Edges <- 0.5 * Z %*% A %*% t(Z)
.ones <- matrix(1,n,n)
diag(.ones) <- 0
Tot <- Z %*% .ones %*% t(Z)/2
Theta <- Edges / Tot
```

- $\mathbf{Z}$: 2 clusters $\times$ 34 vertices
- Θ : edge probabilities

Matrix factorization view: $A \approx Z^T \Theta Z$



Reordering by cluster reveals the block structure



- Sorting vertices by cluster membership makes the block structure visible
- Each block shows edge density within/between groups

How do we know clustering is good enough?

Finding and evaluating community structure in networks

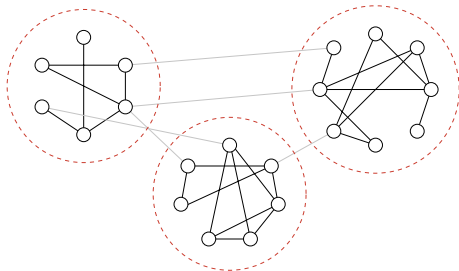
M. E. J. Newman^{1,2} and M. Girvan^{2,3}

¹*Department of Physics and Center for the Study of Complex Systems,
University of Michigan, Ann Arbor, MI 48109-1120*

²*Santa Fe Institute, 1399 Hyde Park Road, Santa Fe, NM 87501*

³*Department of Physics, Cornell University, Ithaca, NY 14853-2501*

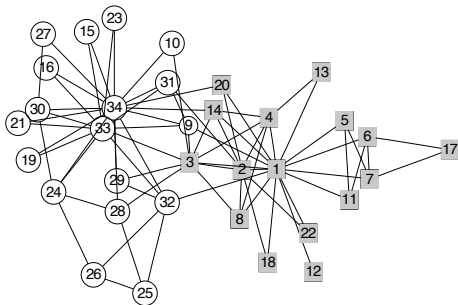
We propose and study a set of algorithms for discovering community structure in networks—natural divisions of network nodes into densely connected subgroups. Our algorithms all share two definitive features: first, they involve iterative removal of edges from the network to split it into communities, the edges removed being identified using one of a number of possible “betweenness” measures, and second, these measures are, crucially, recalculated after each removal. We also propose a measure for the strength of the community structure found by our algorithms, which gives us an objective metric for choosing the number of communities into which a network should be divided. We demonstrate that our algorithms are highly effective at discovering community structure in both computer-generated and real-world network data, and show how they can be used to shed light on the sometimes dauntingly complex structure of networked systems.



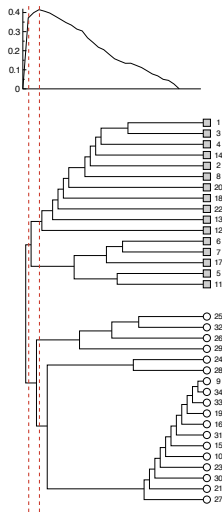
It's not always as perfect as this. . .

- cluster $\approx$ densely connected subgraph
- but how dense?

In a dendrogram of clusters, where should we cut?



This is perhaps what happened after the split. . .



- What will be the proper score?
- The underlying combinatorial optimization is one of the hardest CS problems.

What is modularity? (Newman and Girvan 2004)

Given a partition of vertices into groups, **modularity** Q measures how many more edges fall *within* groups than expected by chance:

$$Q = \frac{1}{2m} \sum_{ij} \left[\mathbf{A}_{ij} - \frac{d_i d_j}{2m} \right] \delta(c_i, c_j)$$

- A_{ij} : adjacency matrix, d_i : degree of vertex i , m : total edges
- $\delta(c_i, c_j) = 1$ if vertices i, j are in the same group
- $Q \in [-0.5, 1]$; higher = stronger community structure
- The null model: edges are distributed proportionally to degree

Why $d_i d_j / 2m$?

- Each vertex i has d_i “stubs” (half-edges). Total stubs: $\sum_i d_i = 2m$.
- Pick a stub from i . It connects to one of $2m - 1 \approx 2m$ other stubs.
- Vertex j has d_j stubs, so:

$$\Pr(\text{stub from } i \text{ lands on } j) = \frac{d_j}{2m}$$

- Vertex i has d_i stubs, so the expected number of edges between i and j :

$$\mathbb{E}[A_{ij}] = \frac{d_i d_j}{2m}$$

Louvain: Greedy optimization of the modularity score



Fast unfolding of communities in large networks

Vincent D Blondel¹, Jean-Loup Guillaume^{1,2},
Renaud Lambiotte^{1,3} and Etienne Lefebvre¹

¹ Department of Mathematical Engineering, Université Catholique de Louvain,
4 avenue Georges Lemaitre, B-1348 Louvain-la-Neuve, Belgium

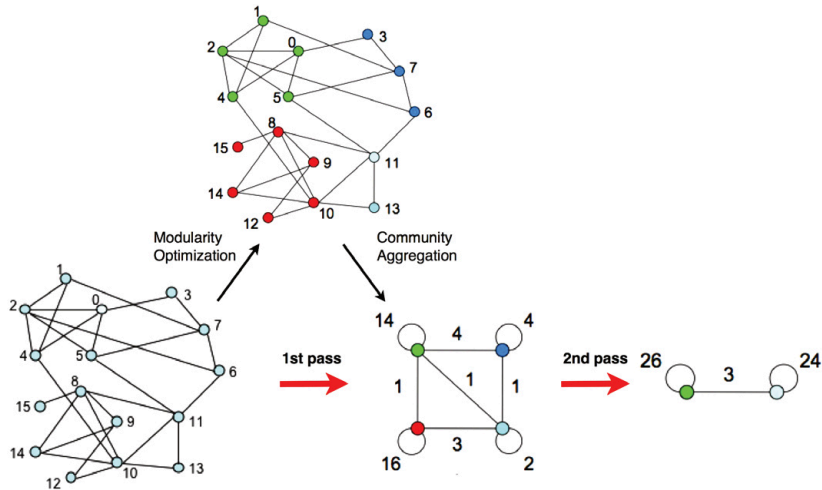
² LIP6, Université Pierre et Marie Curie, 4 place Jussieu, F-75005 Paris, France

³ Institute for Mathematical Sciences, Imperial College London,
53 Prince's Gate, South Kensington Campus, London SW7 2PG, UK
E-mail: vincent.blondel@uclouvain.be, jean-loup.guillaume@lip6.fr,
r.lambiotte@imperial.ac.uk and pixetus@hotmail.com

(Blondel et al. 2008):

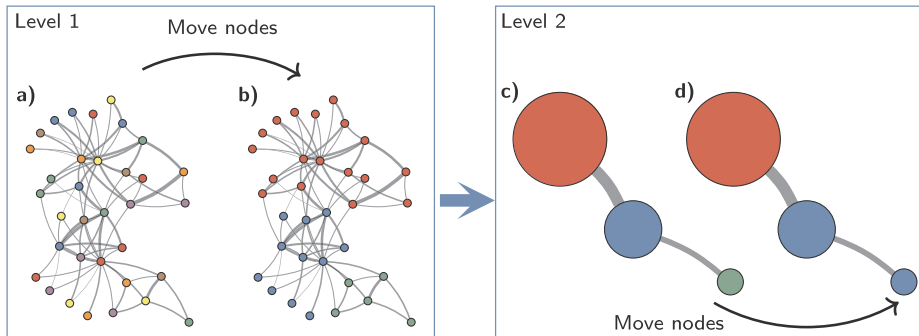
- 1 Each vertex starts as its own community
- 2 Move vertices to neighbour's community if it increases Q
- 3 Aggregate communities into super-nodes
- 4 Repeat until Q converges

Louvain algorithm: vertices $\rightarrow$ “super” nodes



(Blondel et al. 2008)

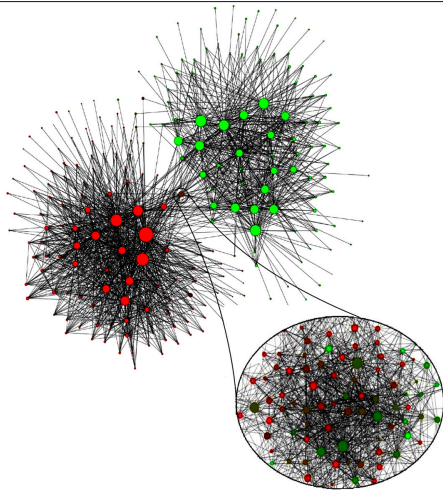
Louvain deals with the “colours” of “super” nodes



Key idea: Keep coarsening a large graph into a smaller one

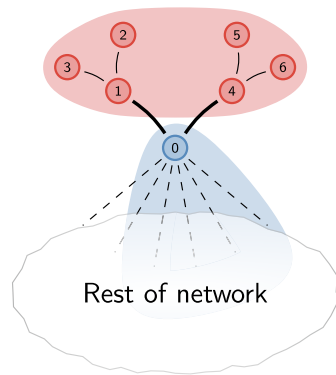
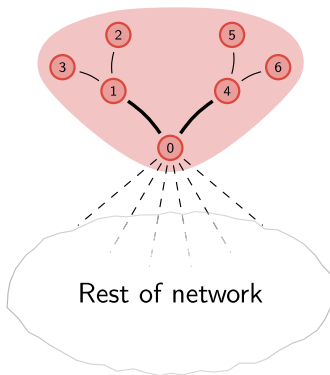
(Traag et al. 2019)

Louvain can easily scale up to a million vertices



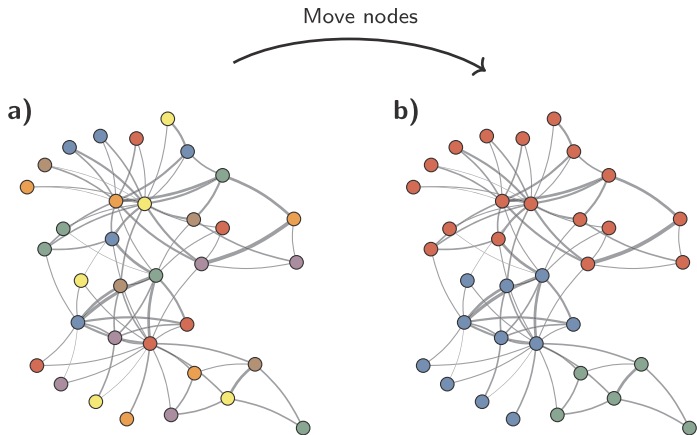
(Blondel et al. 2008)

Louvain's supernode with disconnected components



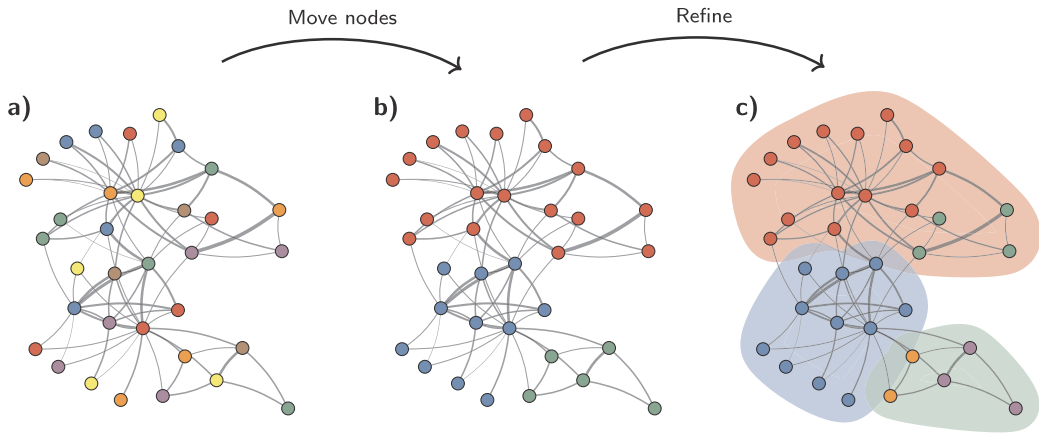
(Traag et al. 2019)

Leiden “refines” to ensure connected within a cluster



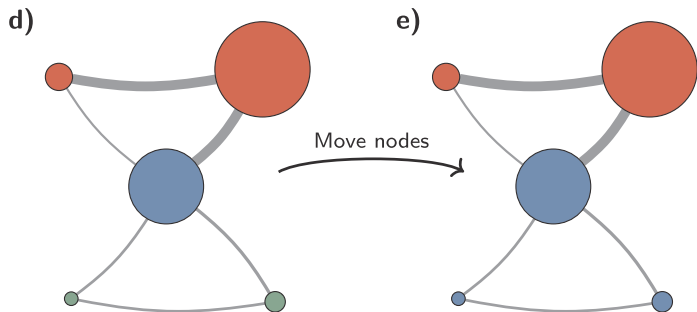
(Traag et al. 2019)

Leiden “refines” to ensure connected within a cluster



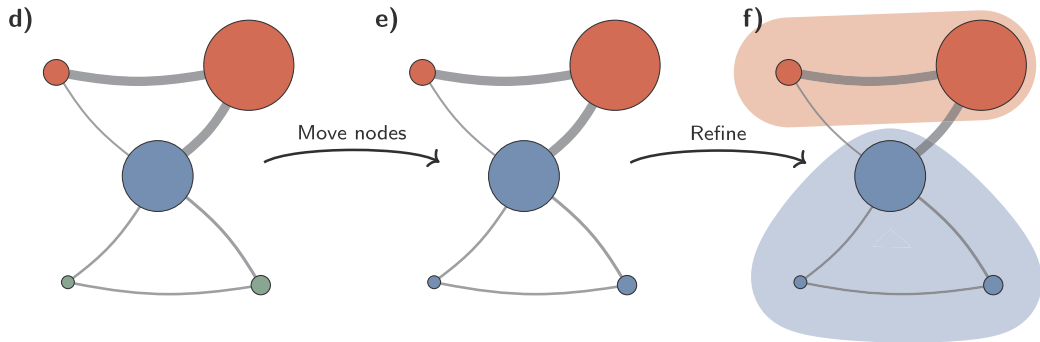
(Traag et al. 2019)

Leiden “refines” to ensure connected within a cluster



(Traag et al. 2019)

Leiden “refines” to ensure connected within a cluster



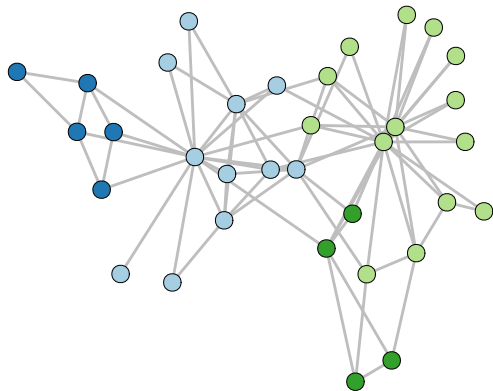
(Traag et al. 2019)

Leiden vs. Louvain

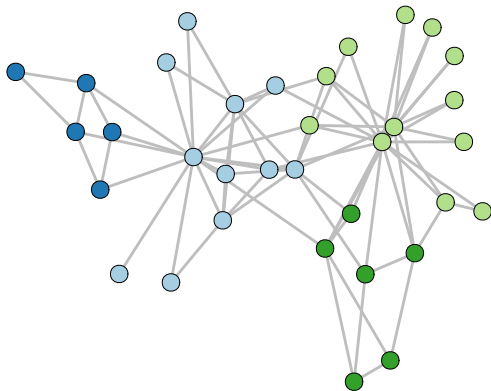
```
cl.louvain <- cluster_louvain(karate)
cl.leiden  <- cluster_leiden(karate,
                             objective_function = "modularity")
```


Leiden vs. Louvain

Louvain Q=0.415



Leiden Q=0.42



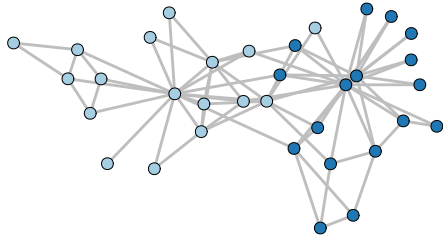
What should we learn from Louvain/Leiden?

When n is large. . .

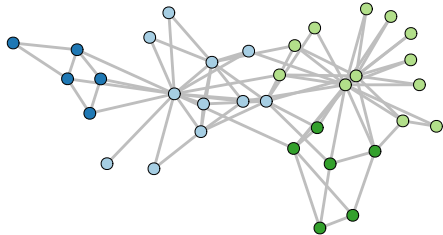
- What shall we do?
- Any counter examples for Leiden?

Side note: The resolution parameter matters

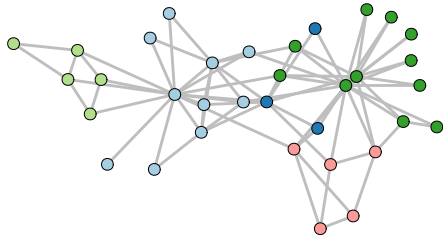
res=0.5 (k=2)



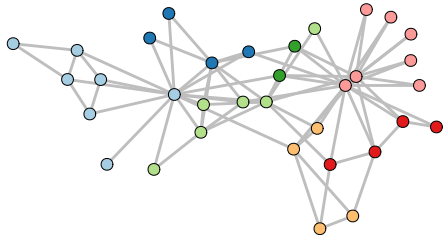
res=1 (k=4)



res=1.5 (k=5)



res=2 (k=7)

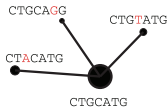


Today's lecture

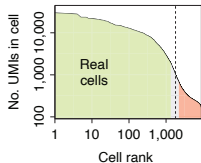
- 1 Graph-based clustering
- 2 Single-cell clustering in practice**
- 3 When clustering goes wrong
- 4 Discussion: How should we test cell-cell communication?
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Recall: the standard single-cell workflow

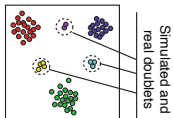
Alignment and molecular counting



Cell filtering and quality control



Doublet scoring

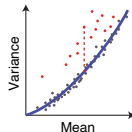


Cell size estimation

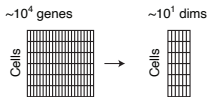
Cells	c_1	c_2	c_3
Gene ₁	2	4	20
Gene ₂	1	2	10
Gene ₃	3	6	30

Cell depth: 6 | 12 | 60

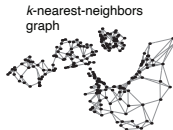
Gene variance analysis



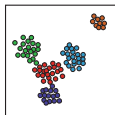
Reduction to a medium-dimensional space



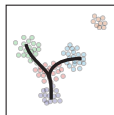
Manifold representation



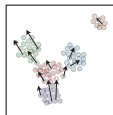
Clustering and differential expression



Trajectories



Velocity estimation



(Kharchenko 2021)

Clustering the PDAC data

```
pdac <- Read10X(data.dir = data.dir)
pdac <- CreateSeuratObject(counts = pdac,
  project = "PDAC_TISSUE_1",
  min.cells = 3, min.features = 200)

pdac <- pdac |>
  NormalizeData() |>
  FindVariableFeatures(nfeatures = 2000) |>
  ScaleData() |>
  RunPCA(npcs = 30) |>
  FindNeighbors(dims = 1:20) |>
  RunUMAP(dims = 1:20)
```

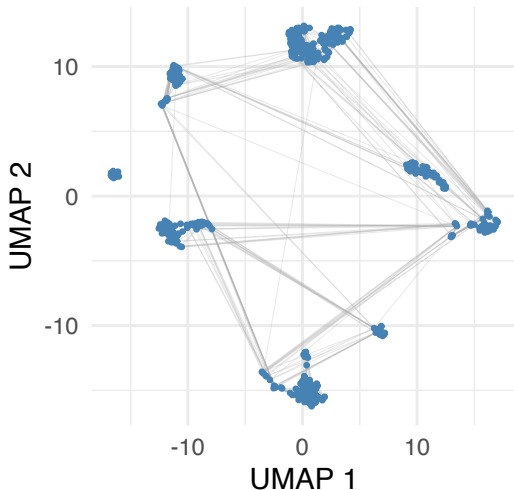
- Same PDAC Tissue 1 dataset from Lecture 8 (Steele et al. 2020)
- FindNeighbors() builds the **KNN** and **SNN** graphs in PCA space

Building the KNN graph

```
knn.mat <- pdac@graphs$RNA_nn  
dim(knn.mat)
```

```
## [1] 995 995
```

- `FindNeighbors(dims = 1:20)` connects each cell to its $k = 20$ nearest neighbours in PCA space
- $A_{ij} = 1$ if cell j is among k -NN of cell i

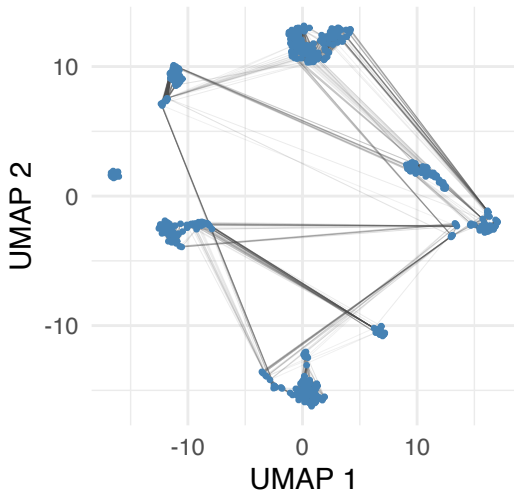


Shared neighbours graph

- **SNN**: weight edges by shared neighbour overlap (Jaccard index)

$$w_{ij} = \frac{|N_i \cap N_j|}{|N_i \cup N_j|}$$

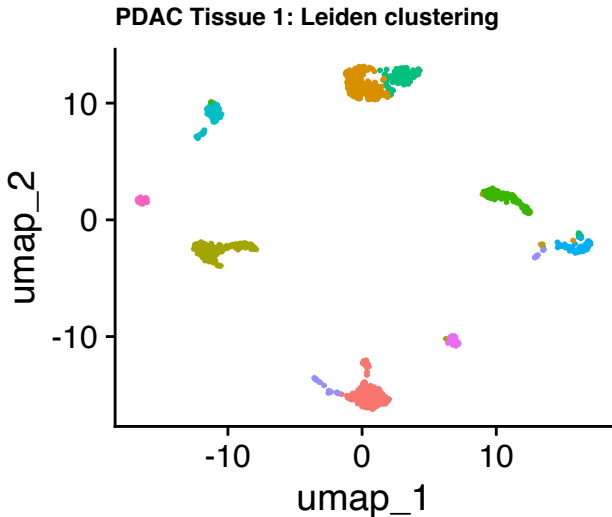
- Cells sharing many neighbours get stronger connections
- This is what Leiden/Louvain clusters on



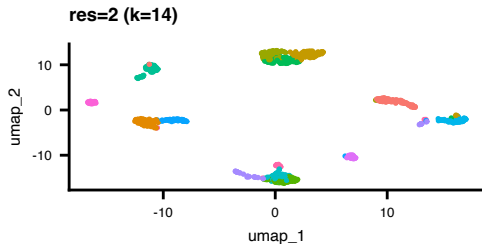
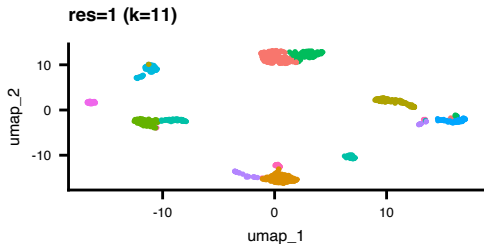
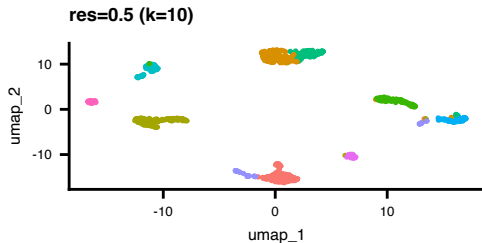
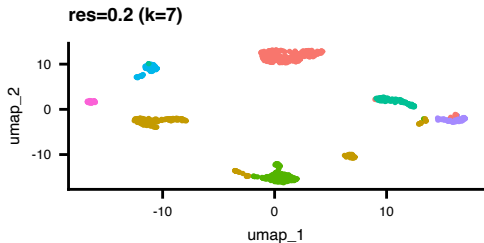
Clustering the PDAC data

```
pdac <- FindClusters(  
  pdac,  
  algorithm = 4,  
  resolution = 0.5)
```

- Leiden on the SNN graph
- 10 clusters found
- Resolution = 0.5



Effect of the resolution parameter



The resolution parameter in the modularity score

The **resolution parameter** γ scales the null model:

$$Q_\gamma = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \gamma \frac{d_i d_j}{2m} \right] \delta(c_i, c_j)$$

- $\gamma > 1$: penalizes large clusters more $\rightarrow$ **more, smaller** clusters
- $\gamma < 1$: tolerates sparser communities $\rightarrow$ **fewer, larger** clusters
- $\gamma = 1$: standard modularity

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Are clusters real or artifacts of the algorithm?

nature methods

Article

<https://doi.org/10.1038/s41592-023-01933-9>

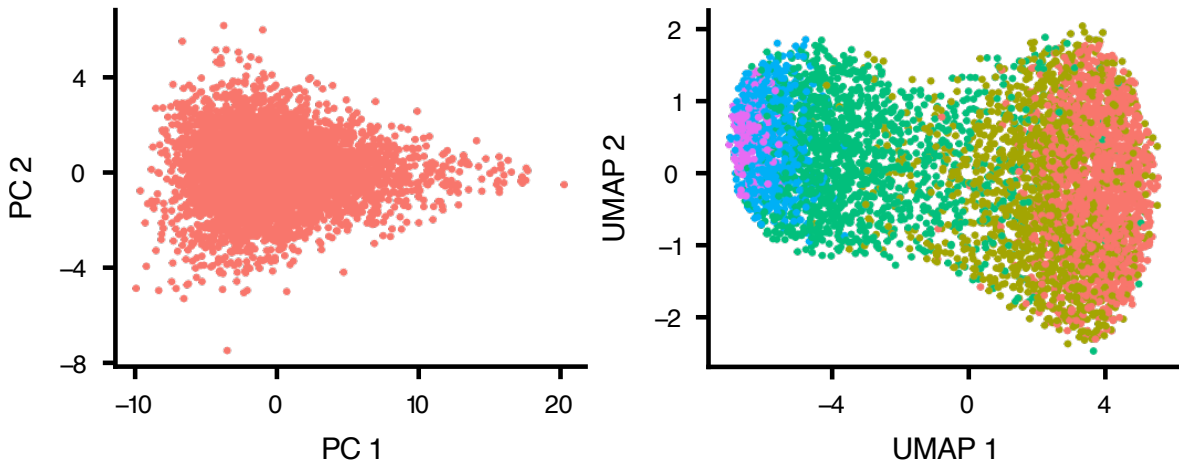
Significance analysis for clustering with single-cell RNA-sequencing data

Received: 1 August 2022

Isabella N. Grabski¹✉, Kelly Street²✉ & Rafael A. Irizarry³✉

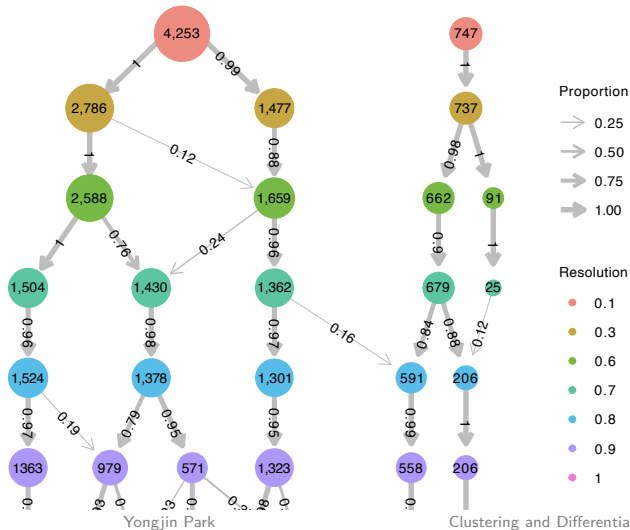
(Grabski et al. 2023)

Clustering a single population finds spurious clusters



Simulated data with **no true subpopulations**. PCA (left) shows one blob; Leiden (right) still finds clusters. (Grabski et al. 2023)

Higher resolution keeps splitting — but are the splits real?

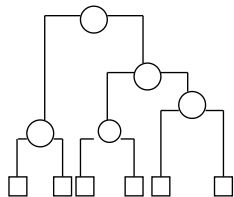


Each node = a cluster at a given resolution. Arrows show how cells redistribute as resolution increases.

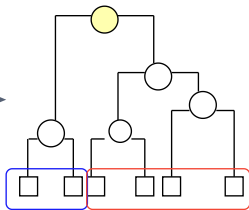
Thin arrows = small proportion of cells moving → potentially spurious splits.

sc-SHC: recursively test each split for significance

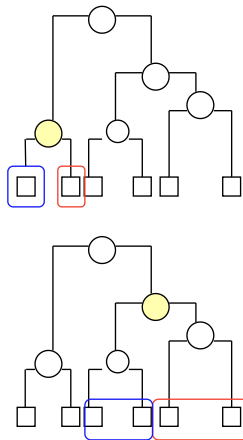
Hierarchically cluster cells
(or pre-computed clusters)



Test split at the root

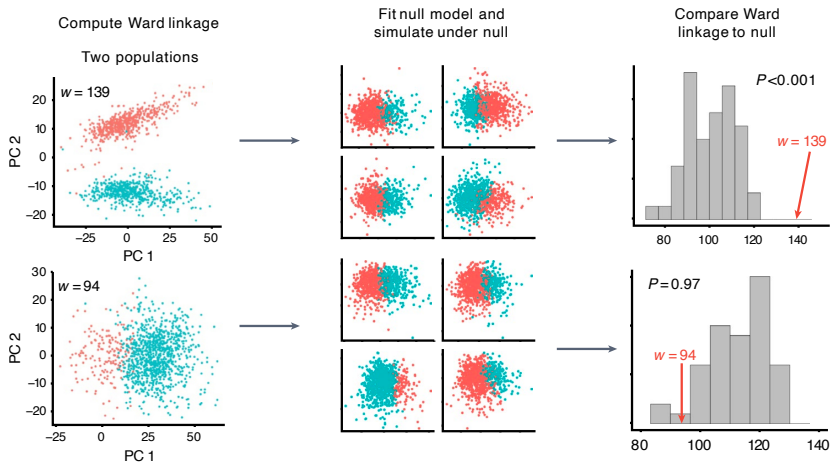


Stop if fail to reject; otherwise
proceed recursively



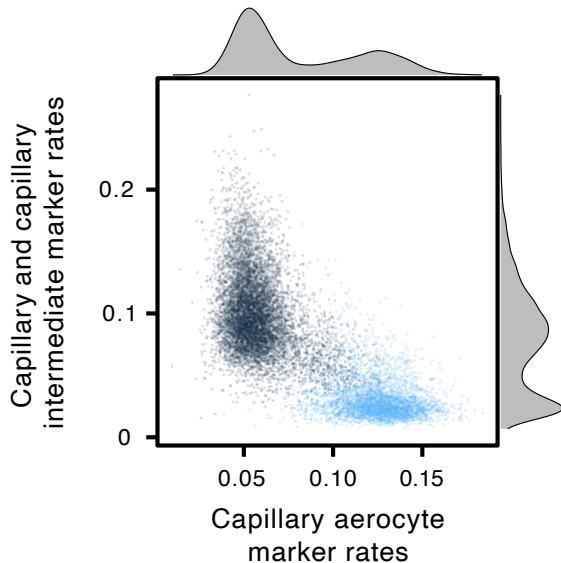
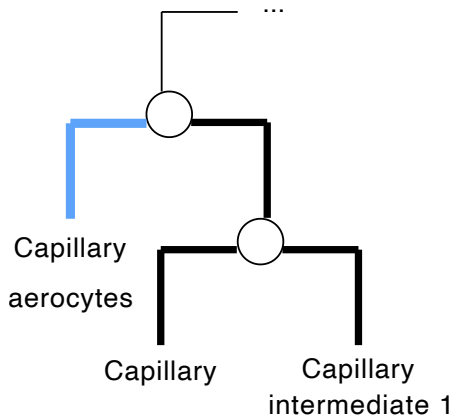
(Grabski et al. 2023)

Can we compute the p-value of each split?

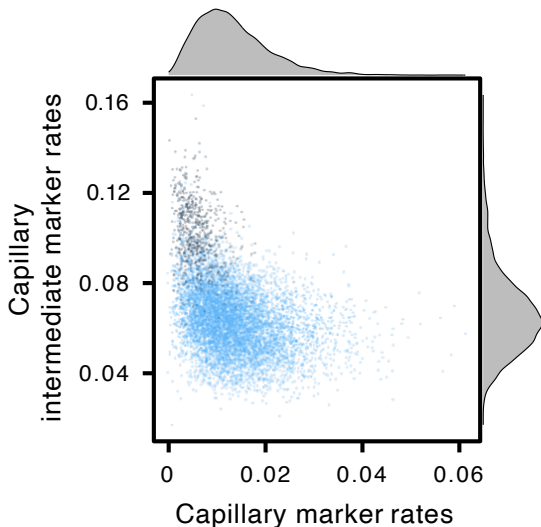
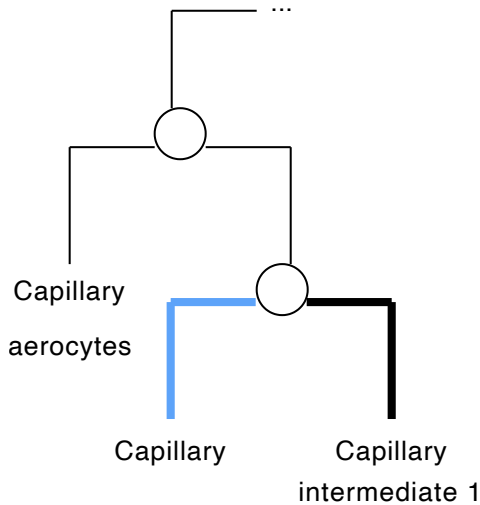


Top: two real populations $\rightarrow$ observed w far from null ($P < 0.001$). Bottom: one population $\rightarrow$ w typical under null ($P = 0.97$). (Grabski et al. 2023)

Capillary aerocytes are a distinct subtype



Not all splits are real: capillary subtypes overlap



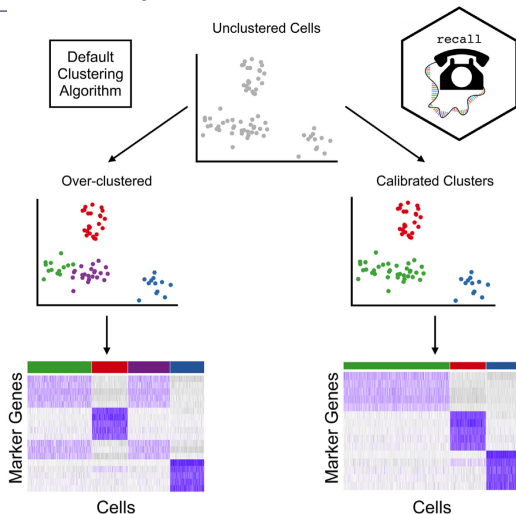
Over-clustering (DenAdel et al. 2025)

Problem: Leiden with a high resolution parameter $\rightarrow$ too many clusters $\rightarrow$ spurious DE genes

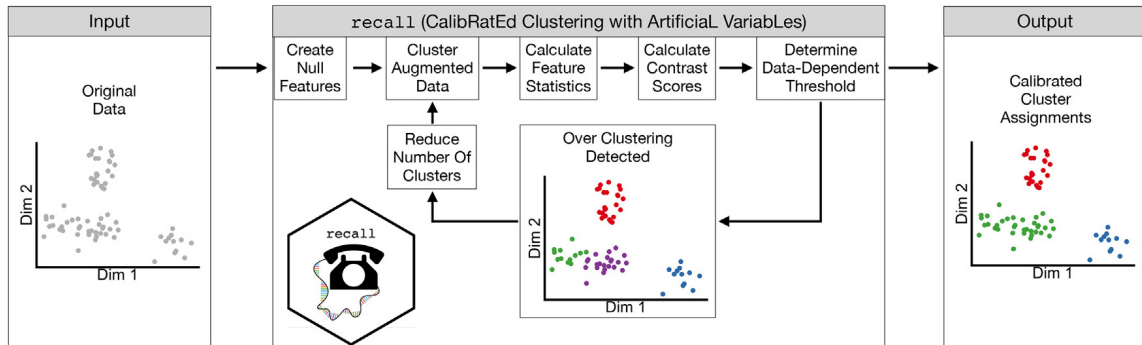
RECALL (calibrated clustering with artificial variables):

- ➊ **Augment** the expression matrix with **artificial null genes** — synthetic variables that match the data distribution but have no association with cell types
- ➋ **Cluster** cells using both real and artificial genes
- ➌ **Test for DE** between clusters, for both real and artificial genes
- ➍ Since artificial genes are known nulls, their test statistics reveal the **double-dipping effect**
- ➎ If real genes are not more significant than artificial ones $\rightarrow$ **merge** those clusters
- ➏ Repeat until clusters are well-calibrated

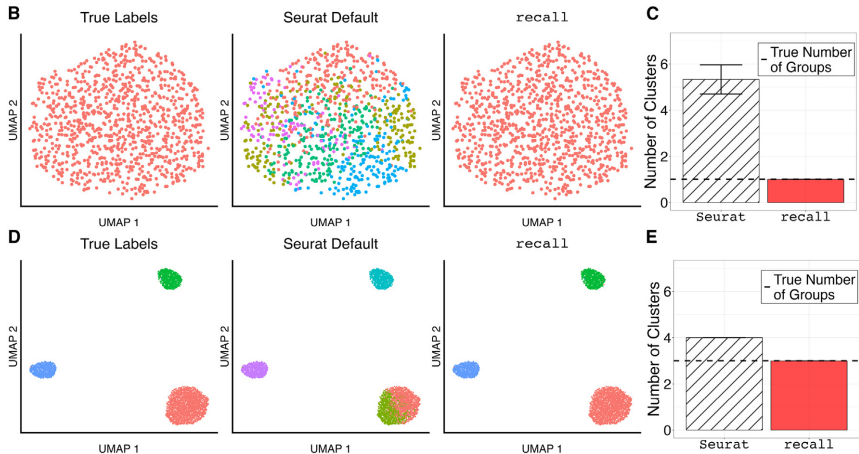
Default clustering over-clusters; RECALL calibrates (DenAdel et al. 2025)



RECALL: iteratively reduce resolution until calibrated

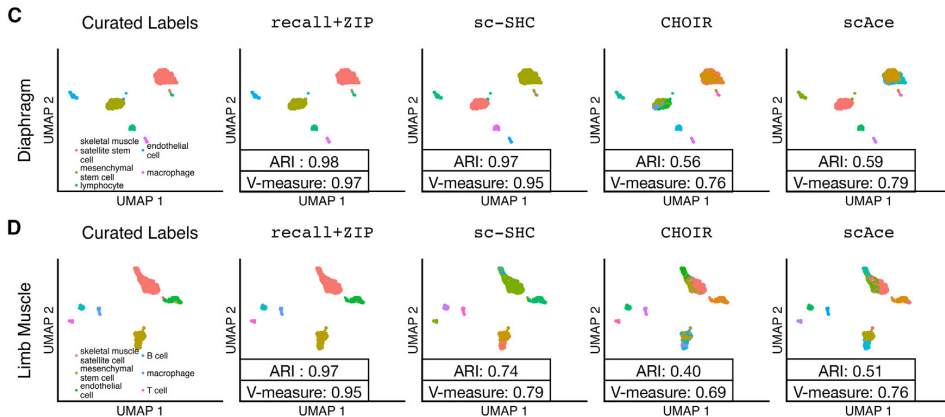


RECALL correctly identifies true clusters



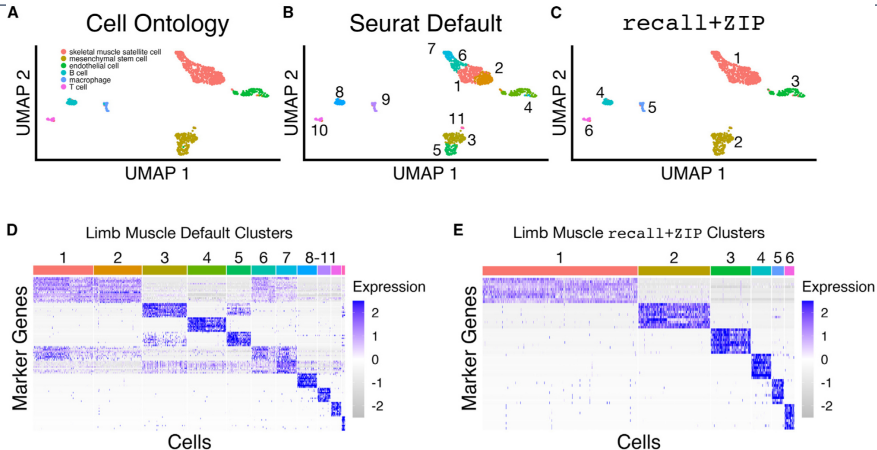
B–C: one population — Seurat finds about 5, RECALL finds 1. D–E: three populations — both recover 3. (DenAdel et al. 2025)

RECALL outperforms alternatives on real tissue data



Diaphragm and limb muscle datasets. ARI and V-measure vs curated labels. (DenAdel et al. 2025)

Seurat default over-splits; RECALL matches cell ontology



Limb muscle: Seurat finds 11 clusters, RECALL finds 6 matching the 6 annotated cell types. Heatmaps show cleaner marker gene blocks. (DenAdel et al. 2025)

How does RECALL generate artificial null genes?

For each gene j , fit a **zero-inflated Poisson** (ZIP) to the observed counts:

$$P(X_j = 0) = \pi_{0j} + (1 - \pi_{0j}) e^{-\lambda_j}, \quad P(X_j = x) = (1 - \pi_{0j}) \frac{\lambda_j^x e^{-\lambda_j}}{x!}$$

Estimate parameters from the data:

- $\hat{\lambda}_j$ from the mean $\bar{x}_j$ and zero-fraction r_{0j} via the Lambert W function
- $\hat{\pi}_{0j} = 1 - \bar{x}_j / \hat{\lambda}_j$

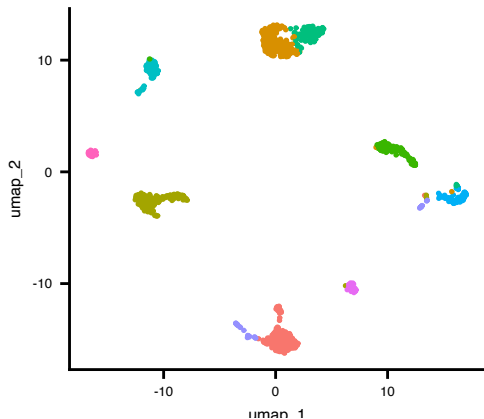
Then sample synthetic null genes: $\tilde{x}_j \sim \text{ZIP}(\hat{\pi}_{0j}, \hat{\lambda}_j)$

Each null gene matches the **marginal** distribution (mean, zero-inflation) of the real gene.

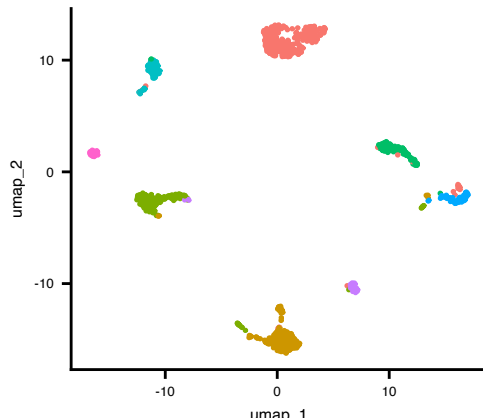
RECALL on the PDAC data

```
library(recall)
pdac.recall <- FindClustersRecall(pdac,
  algorithm = "leiden",
  resolution_start = 0.5, dims = 1:20)
```

Leiden res=0.5 (10 clusters)

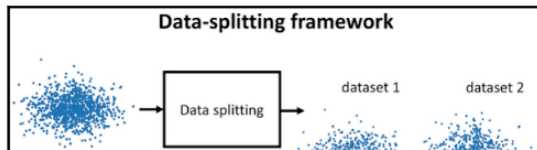


RECALL (8 clusters)



Valid Post-clustering Differential Analysis for Single-Cell RNA-Seq

Graphical Abstract



Authors

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David N. Tse

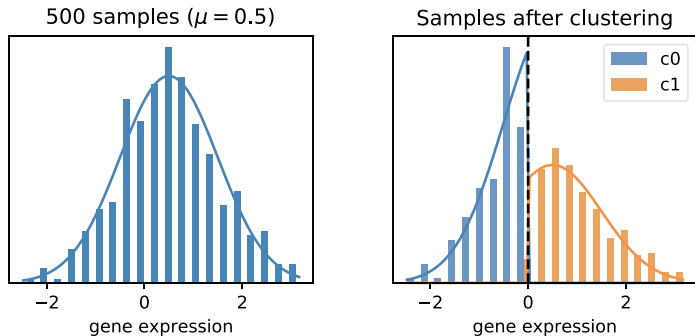
Correspondence

dntse@stanford.edu

- Cluster cells using gene expression → find marker genes that differ between clusters
- But clusters were *defined* to maximize gene expression differences
- This is **selection bias**: the same data used to form and evaluate the hypothesis

(Zhang et al. 2019)

Clustering creates artificial DE even from one population

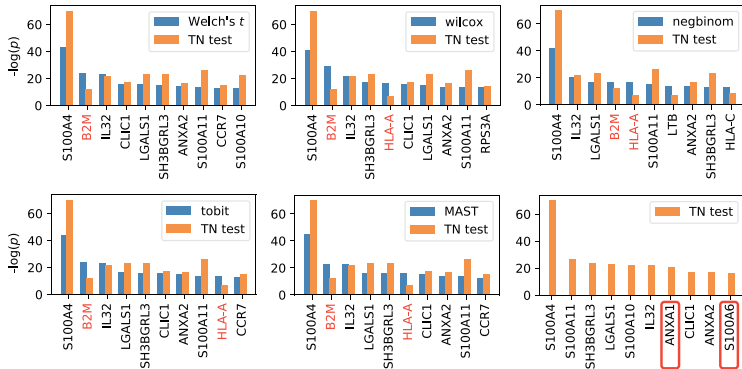
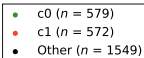
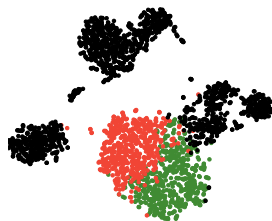


500 cells from **one** distribution ($\mu = 0.5$). After *k*-means, the two clusters look different — but the difference is an artefact. (Zhang et al. 2019)

Standard DE tests are inflated; corrected tests are not

10x PBMC dataset

Most significant genes for Seurat differential expression tests (c0 v c1)



Blue = standard test, orange = truncated normal (TN) corrected test. Red gene names = housekeeping genes that should not be DE. (Zhang et al. 2019)

Clustering then testing → inflated significance

Even when there is **no true grouping**, standard DE after clustering produces many “significant” genes:

- A “double dipping” problem
- The clustering algorithm forces cells into groups
- Cells assigned to the same cluster are more similar *by construction*
- Standard tests (e.g., Wilcoxon, t-test) ignore this selection step → **inflated p-values**

The double-dipping problem

We observe a count matrix X (cells $\times$ genes), where $X_{ij} \sim \text{Poisson}(\lambda_{ij})$

Standard pipeline (double dipping):

- Use X to estimate clusters $\hat{c}_1, \dots, \hat{c}_n$
- Use X again to test: “Is gene j differentially expressed between clusters?”
- The test does not account for the fact that clusters were chosen to make genes look different

Result: under the null (λ_{ij} does not depend on cell type), we still get many small p-values

(Neufeld et al. 2023)

Inference after latent variable estimation for single-cell RNA sequencing data

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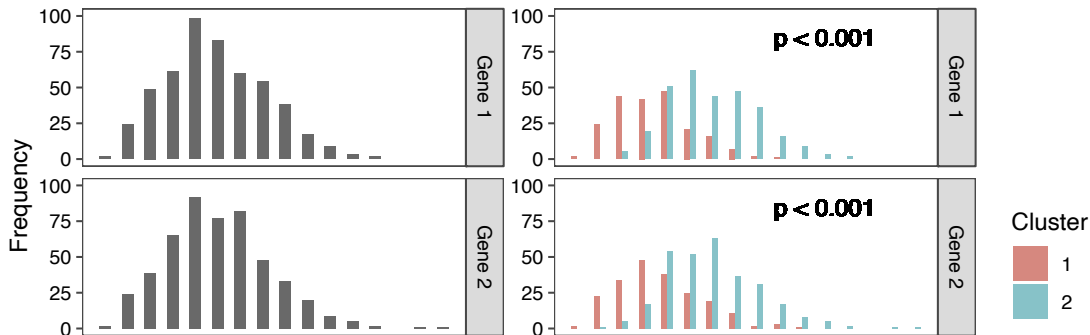
ALEXIS BATTLE

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Department of Computer Science, Johns Hopkins University, Baltimore, MD 21218, USA

DANIELA WITTEN

Department of Statistics, University of Washington, Seattle, WA 98195, USA and Department of
Biostatistics, University of Washington, Seattle, WA 98195, USA

Null genes look DE after clustering: both $p < 0.001$



Two genes drawn from the **same** Poisson distribution across all cells. After clustering (right), both appear significantly DE ($p < 0.001$). (Neufeld et al. 2023)

Count splitting (Neufeld et al. 2023)

Key insight: Poisson thinning creates independent splits

If $X_{ij} \sim \text{Poisson}(\lambda_{ij})$, then for any $\epsilon \in (0, 1)$:

$$X_{ij}^{\text{train}} \mid X_{ij} \sim \text{Binomial}(X_{ij}, \epsilon), \quad X_{ij}^{\text{test}} = X_{ij} - X_{ij}^{\text{train}}$$

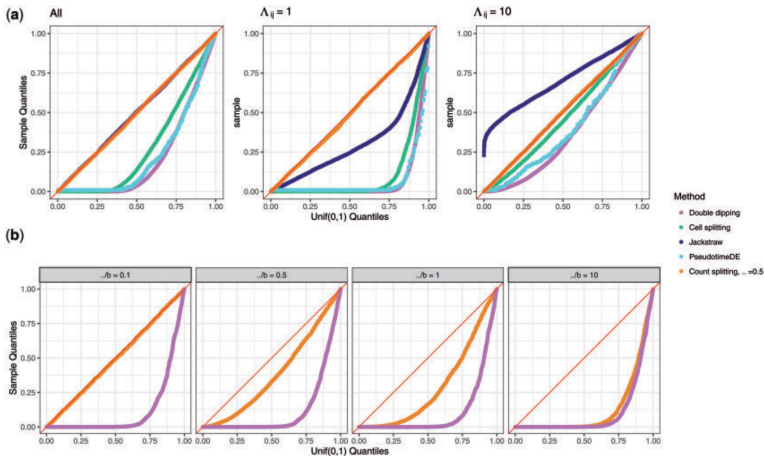
Then $X^{\text{train}} \perp X^{\text{test}}$ (they are **independent!**)

The recipe:

- 1 Split X into X^{train} and X^{test} via Poisson thinning
- 2 Cluster using X^{train} only
- 3 Test for DE using X^{test} only

This controls Type I error because the test data is independent of the clustering.

Count splitting controls Type I error



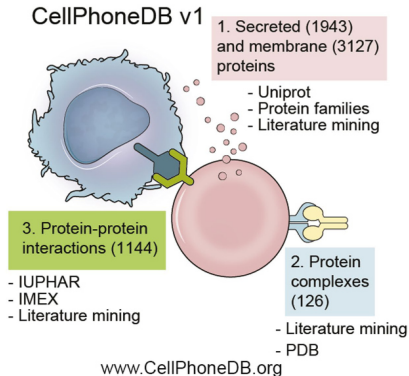
- QQ plots under null: double dipping (purple) inflates p-values
- Count splitting (orange) stays on the diagonal.
- Count splitting across signal-to-noise ratios.

(Neufeld et al. 2023)

Today's lecture

- 1 Graph-based clustering
- 2 Single-cell clustering in practice
- 3 When clustering goes wrong
- 4 Discussion: How should we test cell-cell communication?**
- 5 Appendix

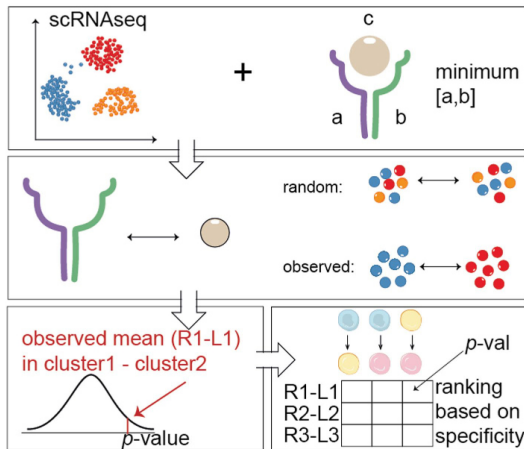
CellPhoneDB infers communication from clusters



- 1 **Cluster** cells into discrete types (e.g., Leiden)
- 2 **Look up** known ligand-receptor (L-R) pairs from a curated database
- 3 For each L-R pair (l, r) and each cell-type pair (A, B):
 - Compute the **mean expression** of l in A and r in B
 - $\text{Score} = \text{mean}(\bar{x}_l^A, \bar{x}_r^B)$
- 4 **Permutation test**: randomly shuffle cell-type labels $\rightarrow$ null distribution of scores
- 5 Report pairs with $p < 0.05$

(Vento-Tormo et al. 2018; Armingol et al. 2021; Efremova et al. 2020)

CellPhoneDB output: significant L-R pairs



(Vento-Tormo et al. 2018)

Discussion

Given what we learned today, what could go wrong with this approach?

Hint 1: Clustering uses the same expression matrix as ...

Hint 2: But L-R genes may not overlap with the genes that ...

Can we fix this? Could count splitting help?

- Split counts: $X = X^{\text{train}} + X^{\text{test}}$
- Cluster on X^{train} , test L-R pairs on X^{test}

But wait . . .

- L-R scores need **mean expression** — do we have enough signal in X^{test} ?
- Ligand in cluster A , receptor in cluster B — we need counts from **different cells**, not the same cell split twice
- What if the clustering itself changes with only half the counts?

Takeaways

Graph-based clustering

- Modularity optimization (Louvain/Leiden) is the workhorse of single-cell clustering
- The resolution parameter controls granularity — there is no single “correct” value
- SBMs offer a principled alternative with automatic model selection

When clustering goes wrong

- Clustering then testing on the same data = **double dipping** → inflated p-values
- **sc-SHC**: build significance testing into the clustering itself
- **Count splitting**: exploit Poisson thinning to create independent train/test sets
- **RECALL**: use artificial null genes to calibrate and detect over-clustering

Today's lecture

- 1 Graph-based clustering
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- 5 Appendix**

No perfect clustering algorithm

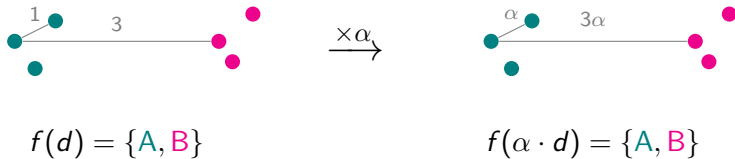
Impossibility Theorem (Kleinberg 2002): no clustering function f can simultaneously satisfy all three axioms:

- ❶ **Scale invariance**: scaling all distances by $\alpha > 0$ doesn't change the clustering
- ❷ **Richness**: every possible partition can be produced by some distance function
- ❸ **Consistency**: shrinking within-cluster distances and expanding between-cluster distances doesn't change the clustering

Any algorithm must violate at least one:

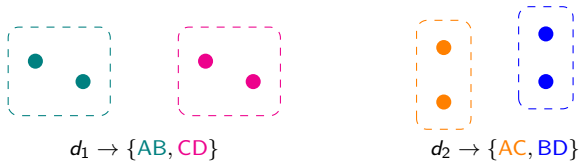
Method	Scale inv.	Richness	Consistency
k -cluster (fixed k)	✓	×	✓
Distance threshold (fixed r)	×	✓	✓
Proportional ($\alpha \cdot d_{\max}$)	✓	✓	×

Axiom 1: Scale invariance



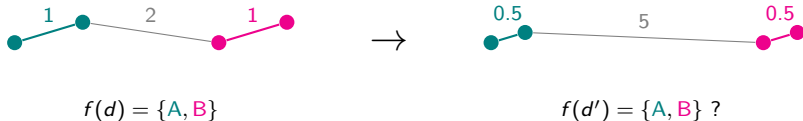
Multiplying all distances by $\alpha > 0$ should not change the partition.

Axiom 2: Richness



For **any** partition of n points, there exists some distance function d that produces it.

Axiom 3: Consistency



If we make clusters “tighter” and “farther apart,” the same partition should remain.

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